BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, BANNU

INTERMEDIATE (12TH CLASS) EXAMINATION, 2021

BIOLOGY

Past Paper 2021

Retyped faithfully from the original board paper — same questions, no changes.

SECTION-A

Time Allowed: 20 Minutes | **Maximum Marks:** 18

1- ہر سوال کے چار ممکنہ جوابات [illegible in source scan] 2- جوابات کی شیٹ (A، B، C، D) کے لیے نیلے یا کالے بال پوائنٹ کا استعمال کریں۔ 3- جواب میں ایک سے زیادہ دائرے بھرنے سے جواب غلط تصور ہوگا۔

Q1. BCG vaccine is used for

(A) cancer
(B) T.B
(C) fever
(D) sinusitis

Q2. Maximum O_2 carrying capacity of 100ml blood

(A) 20ml
(B) 19.4ml
(C) 4.4ml
(D) 97ml

Q3. During infection raised body temperature is helpful because it .....

(A) Kills pathogens
(B) Stimulates defense system
(C) Activates body organ
(D) Increases number of WBC

Q4. ADH acts on which part of nephron in kidney.

(A) distal tubule
(B) proximal tubule
(C) loop of Henle
(D) collecting duct

Q5. Voltage of Resting membrane potential is

(A) +70
(B) -70
(C) +50
(D) -50

Q6. All are signs of diabetes mellitus except one.....

(A) Polyuria
(B) Polydipsia
(C) Polyphagia
(D) Phenylketonuria

Q7. Total number of bones in skull is

(A) 22
(B) 23
(C) 24
(D) 12

Q8. Secondary oocyte after penetration by sperm through meiosis-II gives:

(A) One ovum & second polar body
(B) Two polar bodies
(C) Two ovum
(D) Four polar bodies

Q9. Fraternal twins result from fertilization of ....

(A) One ovum by one sperm
(B) Two ova by two sperms
(C) One ovum by two sperms
(D) Two ova by one sperm

Q10. Random loss of alleles in population is called

(A) mutation
(B) selection
(C) gene flow
(D) genetic drift

Q11. The possible genotype of blood group 'A' will be:

(A) I^AI^A or I^Ai
(B) I^AI^B or I^Ai
(C) I^AI^B
(D) ii

Q12. The Female in butterflies is

(A) XX
(B) XY
(C) XO
(D) X

Q13. Longest bone of the body is

(A) stapes
(B) humerus
(C) ulna
(D) femur

Q14. Nullisomy can be represented as

(A) 2n-2
(B) 2n+2
(C) 2n-1
(D) 2n

Q15. The most efficient nitrogen fixation is .....

(A) Atmospheric
(B) Industrial
(C) Biological
(D) All

Q16. Which one do the function of glue in field of genetic engineering?

(A) Endonuclease
(B) Ligase
(C) Restriction enzyme
(D) Polymerase

Q17. The concept of vaccine is:

(A) To produce antibodies
(B) To kill germs
(C) maintain health
(D) Catalytic

Q18. The total number of vertebrae in human is

(A) 30
(B) 33
(C) 26
(D) 12

Section – B

Maximum Marks: 40

Note: Q-II: Answer any TEN parts. Each part carries FOUR marks.

1. How transport of carbon dioxide takes place in Human blood both as bicarbonate ions and as carboxyhaemoglobin?

2. Write a brief note on structure of Nephron.

3. How osmoregulation process of marine animals is different from fresh water animals?

4. Define Arthritis. Mention its symptoms and causes.

5. What is Reflex Arc? Mention its different components.

6. Mention the three big parts of Hindbrain along with their functions.

7. Write the names of two important hormones of posterior lobe of pituitary gland along with their functions.

8. What do you know about process of spermatogenesis in male human?

9. Mention the role of cytoplasm in process of development by giving example.

10. How many blood types are there in Human? Mention the names of antigen and antibodies of each type.

11. What do you know about semi-conservative model of DNA replication?

12. Write a brief note on Darwin's theory of Evolution.

13. What is gene therapy? What three approaches are used by researchers for correcting faulty gene?

Section – C

Maximum Marks: 27

> Note: Note: Attempt any THREE questions. All questions carry equal marks.

QIII. Q-III

(a) Discuss the osmoregulatory function of kidney.

(b) Write about four groups of vertebrae in Human vertebral column.

QIV. Q-IV

(a) Define nerve impulse. Differentiate between resting membrane potential and active membrane potential.

(b) How is insulin secreted by pancreas and what functions are performed by insulin?

QV. Q-V

State Mendel's Law of Independent Assortment of gene and prove the phenotypic and genotypic ratio with the help of making checker-board.

QVI. Q-VI

(a) Differentiate between aneuploidy and euploidy.

(b) Give argument for using fossil evidence in favour of organic evolution.